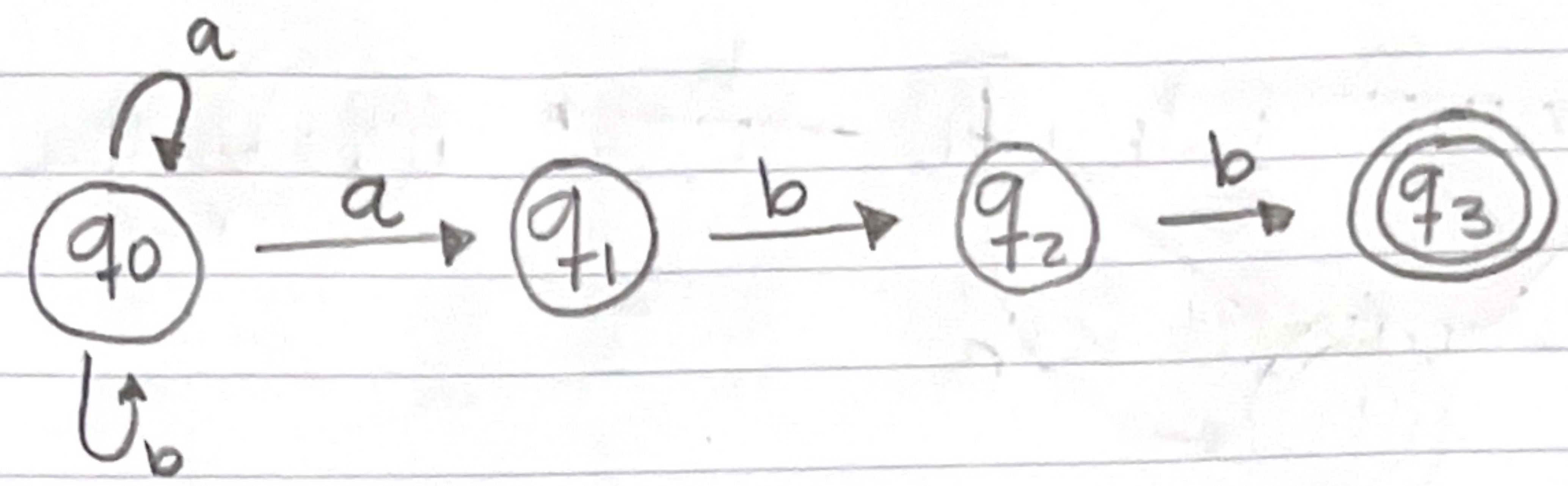
Lab 3:

1).



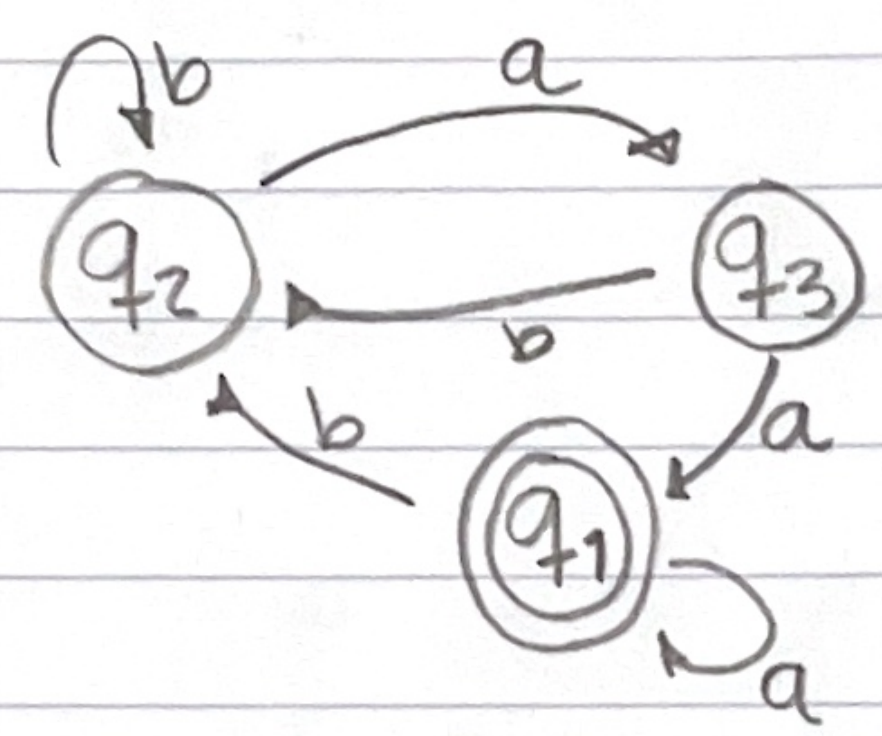
$q_0 = \varepsilon + q_0 a + q_0 b \rightarrow \varepsilon + q_0(a+b) = (a+b)^*$

$q_1 = q_0 a \rightarrow (a+b)^* a$

$q_2 = q_1 b \rightarrow (a+b)^* ab$

$q_3 = q_2 b \rightarrow (a+b)^* abb$ #

2).



$\rightarrow q_1 = a(q_3 + q_1)$

$q_1 = q_1 a + q_3 a$

$q_2 = q_1 b + q_2 b + q_3 b$

$q_2 = b(q_1 + q_2 + q_3)$

$q_3 = q_2 a$

$q_2 = q_1 b + q_2 b + q_2 a$

$q_2 = a(q_3 + q_1)b + q_2(b + ab)$

$q_2 = q_1 b + q_2(b + ab)$

$q_2 = q_1 b(b+ab)^*$

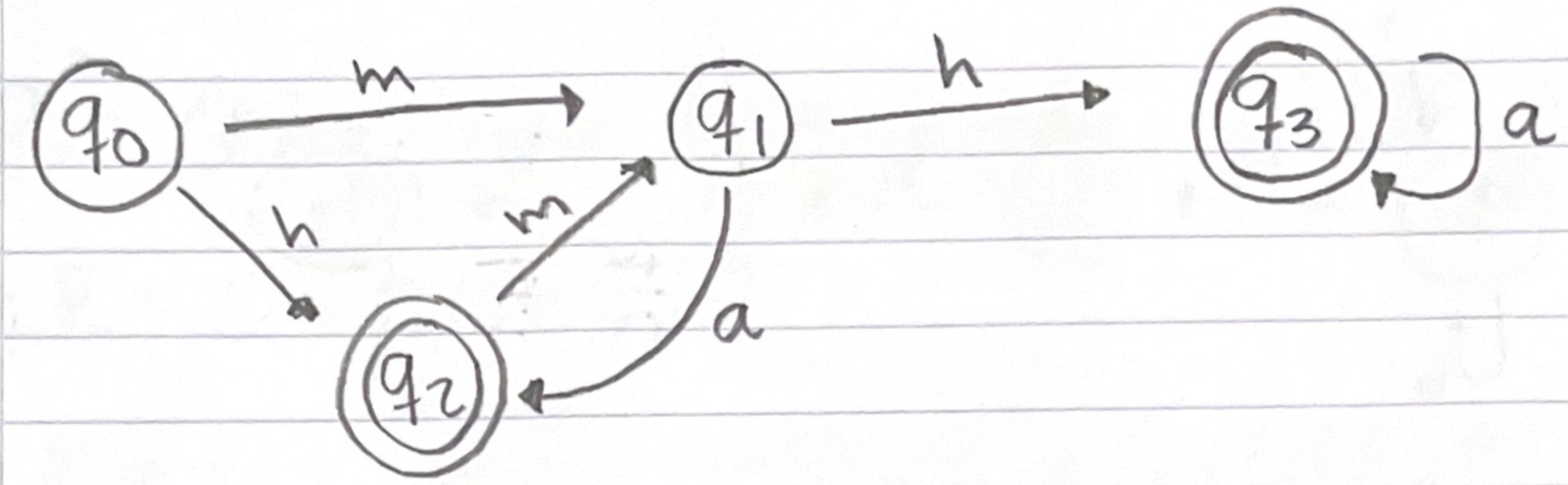
$q_3 = q_1 b(b+ab)^* a$

$q_1 = q_1 a + q_1 b(b+ab)^* aa$

$q_1 = q_1(a + b(b+ab)^* aa)$

$q_1 = (a + b(b+ab)^* aa)^*$ #

3).



$q_0 = \varepsilon$

$q_1 = q_0 m + q_2 m$

$q_2 = q_0 h + q_1 a$

$q_3 = q_1 h + q_3 a$

$q_1 = \varepsilon m + (\varepsilon h + q_1 a) m$

$q_1 = m + (h + q_1 a) m$

$q_1 = m + hm + q_1 am$

$q_1 = m + hm (am)^*$

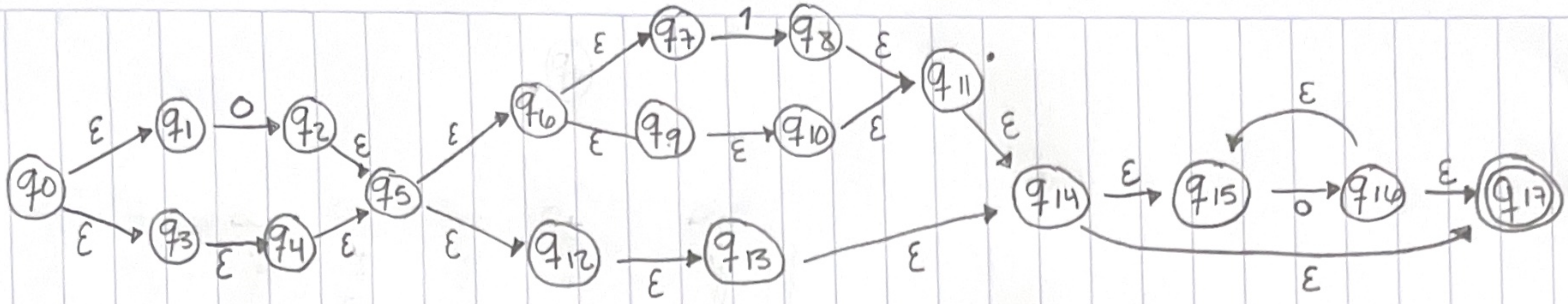
$q_2 = q_0 h + q_1 a$

$q_2 = \varepsilon h + (m + hm (am)^*) a$

$q_3 = q_1 h + q_3 a$

$q_3 = m + hm (am)^* h + q_3 a$

$q_3 = m + hm (am)^* h a^*$ #



$q_0 = \varepsilon$

$q_1 = q_0\varepsilon$

$q_2 = q_1 0$

$q_3 = q_0\varepsilon$

$q_4 = q_3\varepsilon$

$q_5 = q_2\varepsilon + q_4\varepsilon$

$q_6 = q_5\varepsilon$

$q_7 = q_6\varepsilon$

$q_8 = q_7 1$

$q_9 = q_6\varepsilon$

$q_{10} = q_9\varepsilon$

$q_{11} = q_8\varepsilon + q_{10}\varepsilon$

$q_{12} = q_5\varepsilon$

$q_{13} = q_{12}\varepsilon$

$q_{14} = q_{11}\varepsilon + q_{13}\varepsilon$

$q_{15} = q_{14}\varepsilon + q_{16}\varepsilon$

$q_{16} = q_{15} 0$

$q_{17} = q_{16}\varepsilon + q_{14}\varepsilon$

Procedimiento:

$q_2 = \overset{\varepsilon}{q_1} 0 \rightarrow q_3 = \overset{\varepsilon}{q_0} \longrightarrow q_4 = q_3$

$q_5 = q_2\varepsilon + q_4\varepsilon$

$\underline{q_5 = 0 + \varepsilon}$

$q_6 = q_5\varepsilon$

$q_7 = q_6\varepsilon = q_5$

$q_8 = q_7 1 = \underline{q_5 1}$

$q_9 = q_6\varepsilon = q_5$

$q_{10} = q_9\varepsilon = q_5$

$q_{11} = q_8\varepsilon + q_{10}\varepsilon$

$q_{11} = q_5 1 + q_5\varepsilon$

$\underline{q_{11} = q_5(1+\varepsilon)}$

$q_{12} = q_5$

$q_{13} = q_{12}\varepsilon = q_5$

$q_{14} = q_{11}\varepsilon + q_{13}\varepsilon$

$q_{14} = q_5(1+\varepsilon) + q_5$

$\underline{q_{14} = q_5(1+\varepsilon)}$

$q_{15} = q_{14}\varepsilon + q_{16}\varepsilon$

$q_{15} = \underline{q_{14}} + \underline{q_{15} 0}$

$q_{15} = \underline{q_{14} 0^*}$

$q_{16} = q_{15} 0 \rightarrow \underline{q_{16} = q_{14} 0^* 0}$

$q_{17} = q_{16}\varepsilon + q_{14}\varepsilon$

$q_{17} = q_{14} 0^* 0 + q_{14}\varepsilon$

$(\varepsilon + 0)$

$q_{17} = q_{14}(0^* 0 + \varepsilon) \rightarrow q_{14} 0^* \rightarrow q_5 (1+\varepsilon) 0^*$

$\underline{\underline{q_{17} = (0+\varepsilon)(1+\varepsilon) 0^*}}$ //